

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276394

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Jing; Lei et al.

METHOD OF USING PROCESSING OVEN

Abstract

A method of using a processing oven may include disposing at least one substrate in a chamber of the oven and activating a lamp assembly disposed above them to increase their temperature to a first temperature. A chemical vapor may be admitted into the chamber above the at least one substrate and an inert gas may be admitted into the chamber below the at least one substrate. The temperature of the at least one substrate may then be increased to a second temperature higher than the first temperature and then cooled down.

Inventors: Jing; Lei (Santa Clara, CA), Lane; M Ziaul (San Jose, CA), Sautter; Kenneth (Sunnyvale, CA), Song; Kang (San Jose, CA)

Applicant: Yield Engineering Systems, Inc. (Fremont, CA)

Family ID: 83547142

Assignee: Yield Engineering Systems, Inc. (Fremont, CA)

Appl. No.: 19/213254

Filed: May 20, 2025

Related U.S. Application Data

parent US continuation 18387093 20231106 parent-grant-document US 12330228 child US 19213254

parent US continuation 17872320 20220725 parent-grant-document US 11850672 child US 18387093

parent US continuation 17692760 20220311 parent-grant-document US 11465225 child US 17872320

parent US continuation-in-part 17463012 20210831 parent-grant-document US 11296049 child US 17692760

Publication Classification

Int. Cl.: **B23K1/015** (20060101); **B23K1/00** (20060101); **B23K1/008** (20060101); **B23K1/012** (20060101); **B23K3/08** (20060101); **B23K101/40** (20060101); **B23K101/42** (20060101); **H01L21/60** (20060101); **H01L23/00** (20060101)

U.S. Cl.:

CPC **B23K1/015** (20130101); **B23K1/008** (20130101); **B23K3/085** (20130101); **H01L24/742** (20130101); **H01L24/75** (20130101); B23K1/0016 (20130101); B23K1/012 (20130101); B23K2101/40 (20180801); B23K2101/42 (20180801); H01L2021/60007 (20130101); H01L2021/60135 (20130101); H01L2021/6027 (20130101); H01L2224/75272 (20130101); H01L2224/75283 (20130101); H01L2224/75802 (20130101); H01L2224/76272 (20130101); H01L2224/76802 (20130101); H01L2224/77272 (20130101); H01L2224/77802 (20130101); H01L2224/78272 (20130101); H01L2224/78802 (20130101)

Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/387,093, filed Nov. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/872,320, filed Jul. 25, 2022 (now U.S. patent Ser. No. 11/850,672), which is a continuation of U.S. patent application Ser. No. 17/692,760, filed Mar. 11, 2022 (now U.S. Pat. No. 11,465,225), which is a continuation-in-part of U.S. patent application Ser. No. 17/463,012, filed Aug. 31, 2021 (now U.S. Pat. No. 11,296,049), each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a method of using a processing oven.

BACKGROUND

[0003] The drive towards smaller and less expensive electronic devices has driven the development of advanced semiconductor packaging technologies. Semiconductor packaging technologies use solder joints to form several interfaces. For instance, in some electronic packages, wires attached using solder (called wire bonds) or flip-chip solder bumps (called controlled collapse chip or C4 connection) are used to electrically connect (i.e., interconnect) a semiconductor die (or IC chip) to a substrate. In some cases, the die-substrate assembly may also be interconnected to other substrates or interposers (e.g., printed circuit board, etc.) using solder joints or solder bumps (called a ball grid array connection). To form such interconnections, a solder material is reflowed to complete the interconnect.

[0004] Conventionally, solder reflow is accomplished by passing the assembly (die-substrate assembly, wafers, etc.) through a batch reflow oven in which the assembly passes through different zones of the oven on a conveyor belt. Different zones preferably are maintained at different temperatures. As the assembly passes through the different zones, it is heated and cooled such that the assembly is subjected to a reflow temperature profile. A chemical vapor (e.g., a flux vapor) at atmospheric pressure may also be introduced into the oven at one or more zones to assist with the reflow process. Introducing chemical vapor for solder reflow at atmospheric pressure is not optimal for small solder geometries because atmospheric pressure may provide a competing force to the chemical vapor pressure and hinder the vapor from reaching all regions of the solder bumps during reflow. As a result, solder yield may be reduced. As semiconductor technology advances, the solder

bumps are becoming too small for batch thermal reflow systems to achieve desirable solder yield. Small solder bumps on devices require more precise thermal process control and chemical vapor delivery to enable reflow with high yield. The processing ovens and solder reflow methods of the current disclosure may alleviate one or more of the above-described issues.

SUMMARY

[0005] Several embodiments of a solder reflow oven (or a processing oven) and methods of using the oven are disclosed. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only. As such, the scope of the disclosure is not limited solely to the disclosed embodiments. Instead, it is intended to cover such alternatives, modifications, and equivalents within the spirit and scope of the disclosed embodiments. Persons skilled in the art would understand how various changes, substitutions and alterations can be made to the disclosed embodiments without departing from the spirit and scope of the disclosure.

[0006] In one aspect, a method of using a solder reflow oven is disclosed. The method may include disposing at least one substrate including solder in a chamber of the oven. The method may also include decreasing a pressure of the chamber to a first pressure between about 0.1-50 Torr. After decreasing the pressure of the chamber, the temperature of the at least one substrate may be increased to a first temperature. Formic acid vapor may be admitted into the chamber above the at least one substrate while nitrogen is admitted into the chamber below the at least one substrate. The method may also include removing at least a portion of the formic acid vapor from the enclosure. After the removing step, the temperature of the at least one substrate may be further increased to a second temperature higher than the first temperature. The at least one substrate may be maintained at the second temperature for a first time. And then, the at least one substrate may be cooled.

[0007] Various embodiments of the oven may alternatively or additionally include one or more of the following features: increasing the temperature may include increasing the temperature of the at least one substrate using a lamp assembly disposed in the chamber above the at least one substrate; further increasing the temperature may include further increasing the temperature of the at least one substrate using a lamp assembly disposed in the chamber above the at least one substrate; increasing the temperature may include increasing the temperature of the at least one substrate to a temperature between about 150° C.-180° C.; further increasing the temperature may include increasing the temperature of the at least one substrate to a temperature between about 220° C.-250° C.; admitting formic acid vapor into the chamber may include admitting formic acid vapor having a formic acid concentration between about 3-15% into the chamber; admitting formic acid vapor into the chamber may include admitting formic acid vapor into the chamber from a source that is adapted to vary a concentration of formic acid in the formic acid vapor; admitting formic acid vapor into the chamber may include admitting formic acid vapor into the chamber from a bubbler; cooling the at least one substrate may include cooling the at least one substrate to a temperature between about 15° C.-30° C.

[0008] Various embodiments of the oven may alternatively, or additionally, include one or more of the following features: the first time is between about 0.5 minutes to 4 minutes; admitting formic acid vapor into the chamber may include admitting formic acid vapor into the chamber while the at least one substrate is rotating in the chamber; the method may further include lowering the at least one substrate in the chamber to a dosing zone after increasing the temperature of the at least one substrate and before admitting formic acid vapor into the chamber; cooling the at least one substrate may include lowering the at least one substrate in the chamber to contact a cold plate of the chamber; the method may further include directing a liquid coolant through the cold plate to assist in cooling the at least one substrate; cooling the at least one substrate may include lowering the at least one substrate in the chamber to a cool-down zone of the chamber and directing an inert gas on the at least one substrate to assist in cooling the at least one substrate; admitting formic acid vapor into the chamber may include admitting formic acid vapor into the chamber through a

chemical delivery tube positioned proximate a top surface of the chamber; and the chemical delivery tube may comprises one of (i) a straight tube that extends from a side of the chamber towards a center of the chamber, (ii) a tube that includes one or more angled portions, or (iii) a T-shaped tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate exemplary embodiments and, together with the description, are used to explain the disclosed principles. In these drawings, where appropriate, reference numerals illustrating like structures, components, materials, and/or elements in different figures are labeled similarly. It is understood that various combinations of the structures, components, and/or elements, other than those specifically shown, are contemplated and are within the scope of the present disclosure.

[0010] For simplicity and clarity of illustration, the figures depict the general structure of the various described embodiments. Details of well-known components or features may be omitted to avoid obscuring other features, since these omitted features are well-known to those of ordinary skill in the art. Further, elements in the figures are not necessarily drawn to scale. The dimensions of some features may be exaggerated relative to other features to improve understanding of the exemplary embodiments. One skilled in the art would appreciate that the features in the figures are not necessarily drawn to scale and, unless indicated otherwise, should not be viewed as representing proportional relationships between different features in a figure. Additionally, even if it is not specifically mentioned, aspects described with reference to one embodiment or figure may also be applicable to, and may be used with, other embodiments or figures.

[0011] FIGS. 1A and 1B illustrate an exemplary processing oven of the current disclosure;

[0012] FIG. 2 illustrates an exemplary substrate configured to be processed in the processing oven of FIG. 1A;

[0013] FIG. 3 illustrates an exemplary temperature profile that can be used to process the substrate of FIG. 2;

[0014] FIG. 4 illustrates a spindle of the processing oven of FIG. 1A;

[0015] FIGS. 5A and 5B are perspective views of the processing chamber of the processing oven of FIG. 1A;

[0016] FIG. 5C illustrates an exemplary lamp assembly of the processing oven of FIG. 1A;

[0017] FIG. 5D is an exemplary bubbler device that may be used with the processing oven of the current disclosure;

[0018] FIG. 5E are exemplary curves showing the relationship between concentration of formic acid vapor at different bubbler pressures and temperatures;

[0019] FIGS. 6A and 6B are cross-sectional views of portions of the processing chamber of FIG. 5A along a vertical plane;

[0020] FIGS. 7A-7C are top views of the processing chamber of FIG. 5A in different exemplary embodiments;

[0021] FIG. 8 is a flowchart showing an exemplary method of using the processing oven of FIG. 1A;

[0022] FIG. 9 is a flowchart showing another exemplary method of using the processing oven of FIG. 1A;

[0023] FIG. 10 is an exemplary reflow profile that may be applied in the processing oven of FIG. 1A; and

[0024] FIG. 11 is a graph showing pressure and oxygen concentration in the processing oven during an exemplary process.

DETAILED DESCRIPTION

[0025] All relative terms such as “about,” “substantially,” “approximately,” etc., indicate a possible variation of $\pm 10\%$ (unless noted otherwise or another variation is specified). For example, a feature disclosed as being about “t” units long (wide, thick, etc.) may vary in length from (t-0.1t) to (t+0.1t) units. Similarly, a temperature within a range of about 100-150° C. can be any temperature between (100-10%) and (150+10%). In some cases, the specification also provides context to some of the relative terms used. For example, a structure described as being substantially circular or substantially cylindrical may deviate slightly (e.g., 10% variation in diameter at different locations, etc.) from being perfectly circular or cylindrical. Further, a range described as varying from, or between, 5 to 10 (5-10), includes the endpoints (i.e., 5 and 10).

[0026] Unless otherwise defined, all terms of art, notations, and other scientific terms or terminology used herein have the same meaning as is commonly understood by one of ordinary skill in the art to which this disclosure belongs. Some of the components, structures, and/or processes described or referenced herein are well understood and commonly employed using conventional methodology by those skilled in the art. Therefore, these components, structures, and processes will not be described in detail. All patents, applications, published applications and other publications referred to herein as being incorporated by reference are incorporated by reference in their entirety. If a definition or description set forth in this disclosure is contrary to, or otherwise inconsistent with, a definition and/or description in these references, the definition and/or description set forth in this disclosure controls over those in the references that are incorporated by reference. None of the references described or referenced herein is admitted as prior art to the current disclosure.

[0027] FIG. 1A is a side view and FIG. 1B is a perspective view of an exemplary processing oven (oven **100**) of the current disclosure. Processing oven **100** may be used for subjecting any type of samples to any temperature process, such as, for example, solder reflow process. In the discussion below, processing oven **100** will be described in terms of its preferred use as a solder reflow oven. However, this use is exemplary and illustrative only, and is not a requirement. Oven **100** may be used for any other appropriate application.

[0028] Oven **100** includes a process chamber **40**, configured to receive one or more substrates (e.g., wafers, organic/ceramic substrates, semiconductor packages, printed circuit board (PCB), etc.), and adapted to subject the one or more substrates to a high temperature process. For the sake of brevity, in the discussion below, a semiconductor wafer with solder on its top surface (see FIG. 2) will be described as the exemplary substrate that is subject to an exemplary solder reflow process in oven **100**. Reference to a single wafer in the discussion below is only exemplary. In general, any number and type of substrate (as described above) may be processed in oven **100**, and any size of the substrate (e.g., 200 mm wafer, 300 mm wafer, etc.) may be received in chamber **40**.

[0029] FIG. 2 illustrates an exemplary semiconductor wafer **10** that may be received in chamber **40** and subject to a reflow process. As would be recognized by a person skilled in the art, after integrated circuit processing, wafer **10** includes multiple dies (or IC chips), and solder material may be deposited on the I/O pads of the multiple dies in wafer **10**. Prior to dicing the wafer **10** into individual dies, wafer **10** may be subject to a reflow process in oven **100** to form solder bumps **12** thereon (called wafer bumping). Any type of solder material (lead-free solder, lead-tin solder, etc.) may be deposited on wafer **10** and the wafer **10** may be subject to any type of reflow process or reflow profile. Persons having ordinary skill in the art would recognize that reflow process that wafer **10** is subjected can depend on the type of solder material deposited on wafer **10**. Suitable reflow profiles for different types of solder materials are available in the published literature and may be obtained from solder vendors.

[0030] FIG. 3 illustrates an exemplary reflow temperature profile that may be applied to wafer **10** in chamber **40**. In FIG. 3, the X-axis shows time in seconds, and the Y-axis shows temperature in degrees Celsius (° C.).

[0031] A robotic manipulator or arm (not shown) can insert wafer **10** into chamber **40** through an inlet port **42** (see FIGS. **1B** and **5B**). Wafer **10** can be disposed on a rotatable spindle **44** within chamber **40**. FIG. **4** illustrates an exemplary spindle **44** positioned within chamber **40** (see FIG. **5B**). Spindle **44** can include a plurality of arms **46A**, **46B**, **46C** (e.g., **3** arms) that can extend radially outward from a central axis **50**. Each arm **46A**, **46B**, **46C** of spindle **44** can include protrusions or nubs **48A**, **48B**, **48C** that projects upwards from the arm. Wafer **10** can rest on the nubs **48A**, **48B**, **48C** of spindle **44**. Oven **100** also includes a motor assembly **20**, configured to support and control the movement of wafer **10** in chamber **40**. Motor assembly **20** can include one or motors configured to rotate spindle **44** (and wafer **10** supported thereon) in the chamber **40** about its central axis **50**. Spindle **44** can be rotated at any speed. Although not a requirement, in some embodiments, the rotational speed of spindle **44** can vary between about 0 to 20 RPM. In some embodiments, motor assembly **20** can include multiple motors. For example, motor can raise and lower the rotating wafer along axis **50** (on spindle **44**) between multiple vertically spaced-apart regions or zones within chamber **40**, and a separate motor can rotate spindle **44**. Oven **100** can also include a lamp assembly **80** with a plurality of lamps **82** (see FIGS. **5B**, **5C**) configured to heat a top surface of wafer **10** in chamber **40**.

[0032] FIGS. **5A** and **5B** illustrate an exemplary processing chamber **40** of oven **100**. As illustrated in these figures, in some embodiments, chamber **40** can have a substantially cylindrical configuration. A substantially cylindrical shape is not required. In general, processing chamber **40** can have any shape. In some embodiments, the chamber **40** can be substantially square, substantially rectangular, or have another shape. In some embodiments, as illustrated in FIG. **5B**, oven **100** can have a hinged lid **70** attached to base **60**. However, a hinged lid is not a requirement, and in some embodiments, oven **100** can include separable lid **70**. Base **60** and lid **70** define an enclosed space or enclosure **66** (see FIG. **5B**) that contains the spindle **44**.

[0033] The diameter (or width) of the interior or chamber **40** (i.e., enclosure **66**) can be such that the wafer **10** can be positioned and rotated (about axis **50**) in enclosure **66**. The size of chamber **40** and enclosure **66** depends upon the application, for example, the size of the wafer that will be processed in chamber **40**. In some embodiments, in an oven **100** and chamber **40** configured to process 300 mm wafers, enclosure **66** can have a diameter of about 450 mm and a height of about 150 mm. The inlet port **42**, through which wafer **10** is inserted into enclosure **66** of chamber **40** may be defined on base **60**. In some embodiments, a valve or a flap can seal the opening of inlet port **42** into enclosure **66** such that enclosure **66** can be pumped down to a low pressure (or pumped up to a high pressure). Base **60** can include one or more gas ports **62**. In some embodiments, process gas can be directed in enclosure **66** through gas port **62**. In some embodiments, vacuum pump may be coupled to gas port **62** to generate a vacuum in enclosure **66**.

[0034] In some embodiments, as schematically illustrated in FIG. **1A**, a bubbler **300** can be fluidly coupled to gas port **62** such that a process gas, such as a chemical vapor, generated by bubbler **300** can be directed into the enclosure **66** through this port. FIG. **5D** is a schematic illustration of an exemplary bubbler **300**. Bubbler **300** includes an enclosed container **310** containing a liquid chemical **320**. A gas (e.g., an inert gas, such as, for example, nitrogen) is directed into the chemical **320** through an inlet tube **330**. As illustrated in FIG. **5D**, the outlet opening of inlet tube **330** can be submerged in the chemical **320** stored in the bubbler **300** such that the gas exiting the inlet tube **330** passes through (or bubbles through) the chemical **320**. As a result of the gas passing through the chemical **320**, a vapor containing the chemical **320** and the gas (chemical vapor) may collect in a space **340** above the level of the chemical **320** in the container **310**. An outlet tube **350** can be positioned in bubbler **300** such that one end of this tube **350** (the end with its inlet opening) is disposed in the space **340** above the chemical **320**. The opposite end (the end with the outlet opening) of the outlet tube **350** can be fluidly connected to the gas port **62** of the processing chamber **40** (see FIG. **5B**). The temperature and pressure in the container **310** can be controlled to control (or tune) the concentration of the chemical in the chemical vapor and the partial pressure of

the chemical vapor.

[0035] In some embodiments, formic acid liquid can be stored as the chemical **320** in the bubbler **300**, and nitrogen gas may be directed in through the inlet tube **330** to flow through formic acid liquid. As a result, a vapor containing nitrogen gas and formic acid (or formic acid vapor) can be collected in the space **340** above the liquid in container **310**. This vapor mixture can be directed into enclosure **66** of processing chamber **40** through outlet tube **350** and gas port **42**. The concentration of formic acid in the formic acid vapor exiting the bubbler **300** can be varied by controlling the temperature and pressure of the bubbler **300** (e.g., of the liquid formic acid in container **310** of bubbler **300**). Alternatively, or additionally, in some embodiments, the concentration of formic acid in the vapor can be varied (or tuned) by controlling the temperature and pressure of the nitrogen gas directed into the bubbler **300**.

[0036] FIG. 5E is an exemplary graph **500** that shows the concentration of the formic acid in the vapor at different bubbler temperatures and pressures. With reference to this graph, when the bubbler pressure is about 200 Torr and the bubbler temperature is about 45° C., the concentration of formic acid in the formic acid vapor directed to the processing chamber **40** will be about 50%. And when the bubbler temperature is decreased to about 25° C., the concentration of formic acid in the vapor decreases to about 20%. The concentration of the formic acid in the formic acid vapor directed to the processing chamber **40** of oven **100** may be controlled in this manner by varying the bubbler temperature and/or bubbler pressure. The graph illustrated in FIG. 5E is only exemplary.

[0037] With reference to FIG. 5B, in some embodiments, lamps **82** of lamp assembly **80** can be disposed on an underside of lid **70** such that, when activated, the lamps **72** heat a top surface of wafer **10** positioned in enclosure **66**. FIG. 5C illustrates an exemplary lamp assembly **80** on lid **70**. In general, lamp assembly **80** can include any number and any type of lamps **82**. In some embodiments, lamp assembly **80** can include 4-10 infrared (IR) lamps **82** having a power between about 1-10 kW for each lamp, or between about 1.5-3 kW (or about 2 kW) for each lamp **82**. In some embodiments, as illustrated in FIG. 5C, lamp assembly **80** can include seven halogen lamps **82**. Lamps **82** can be arranged (e.g., spaced apart) such that they evenly heat the top surface of the wafer **10** in chamber **40**. In some embodiments, as illustrated in FIG. 5C, the lamps **82** can be arranged closer together at the edges (of lamp assembly **80**) than at the center. As a result of edge effects, the wafer **10** in the chamber **40** may cool down quicker at the edges than at the center. A smaller relative spacing of the lamps **82** at the edges than at the center can assist in achieving an even temperature distribution on the wafer **10**.

[0038] The spacing between the lamps **82** of lamp assembly **80** can depend on the application (e.g., size of the wafer and the oven **100**). With reference to FIG. 5C, in some embodiments (e.g., in an oven **100** configured to process 300 mm wafers), the spacing between the lamps **82** can be such that a.sub.1 is between about 10-30 mm (e.g., about 20 mm), a.sub.2 can be between about 40-60 mm (e.g., about 50 mm), a.sub.3 can be between about 60-80 mm (e.g., about 75 mm), a.sub.4 can be between about 40-60 mm (e.g., about 50 mm), as can be between about 40-60 mm (e.g., about 50 mm), and a.sub.6 can be between about 40-50 mm (e.g., about 45 mm). The above-described spacing values are only exemplary. In general, the spacing between adjacent lamps **82** at the sides of the lamp assembly **80** can be smaller than that at the center of the lamp assembly **80**. In some embodiments, the spacing between adjacent lamps **82** at the sides of the lamp assembly **80** (e.g., a.sub.1 or a.sub.2 or a.sub.6) can be between about 10-60 mm, and the spacing between adjacent lamps **82** at the center of the lamp assembly **80** (e.g., a.sub.3 or a.sub.4) can be greater than that at the sides and between about 40-80 mm.

[0039] In some embodiments, lamps **82** can be controlled by control system **200** (schematically illustrated in FIG. 1A) of oven **100** to heat the wafer **10** at a selected temperature ramp rate. For example, a different number of lamps **82** can be activated to increase or decrease the ramp rate. Alternatively, or additionally, in some embodiments, the power of lamps **82** can be varied to vary the temperature ramp rate in enclosure **66**. In some embodiments, the control system **200** can

control the power to the lamps **82** based on the detected temperature in the chamber **40** (e.g., using a feedback loop). For example, when thermocouples (or pyrometers or another temperature detection sensor) in chamber **40** indicate that the temperature of the wafer is below a desired value, the control system **200** can increase the power to the lamps **82**. In some embodiments, when the thermocouples indicate that the temperature variation in the wafer is above a threshold value (e.g., temperature at the edges of the wafer is below the temperature at the center, etc.), the control system **200** can vary the power to different lamps **82** in the lamp assembly **80**. Although not visible in FIG. 5B, in some embodiments, a window made of a material that is transparent to the wavelength of light emanating from lamps **82** may be provided between base **60** and lid **70** to seal the top of enclosure **66** from the external environment. In such embodiments, the lamps **82** heat the wafer **10** in enclosure **66** through the transparent window. In some embodiments, the window can be made of, for example, quartz, glass, etc.

[0040] FIG. 6A illustrates a side view of chamber **40**. Enclosure **66** of chamber **40** can define a plurality of vertically spaced-apart virtual zones. These can may include cool-down zone **90A** at the base of enclosure **66**, transfer zone **90B** directly above cool-down zone **90A**, dosing zone **90C** directly above transfer zone **90B**, and fast ramp zone **90D** directly above the dosing zone **90C** and at the top of enclosure **66**. These virtual zones **90A-90D** of enclosure **66** can be adapted for different functions during reflow processing of wafer **10**. During processing, spindle **44** is configured to move (i.e., raise and lower) wafer **10** positioned on nubs **48A-48D** (see FIG. 4) between one or more of zones **90A-90D** of enclosure **66**.

[0041] FIG. 6B is a side view of cool-down zone **90A** of enclosure **66**. Cold plate **94** forms the base of enclosure **66**. In some embodiments, while in cool-down zone **90A**, wafer **10** can physically contact, or may be positioned near, cold plate **94** to be cooled by conduction. Cold plate **94** includes coolant channels **94** configured to direct a coolant (water or another liquid coolant) therethrough to remove heat conducted from wafer **10**. Cold plate **94** also includes plurality of openings **95** (see FIG. 7A) that can be configured to direct an inert gas (e.g., nitrogen) into enclosure **66**. In some embodiments, nitrogen can be directed into enclosure **66** via openings **95** to cool underside of wafer **10** by forced convection. A plurality of gas-inlet ports **96** can be positioned in cool-down zone **90A** to direct inert gas (e.g., nitrogen) into enclosure **66**. When wafer is positioned in cool-down zone **90A** (e.g., seated on or positioned proximate cold plate **94**), inert gas (e.g., nitrogen or other suitable inert gas) can be directed into enclosure **66** to flow above wafer **10** and cool top surface of wafer **10** by convection. In some embodiments, gas ports **96** can be disposed on side wall of base **60** (of chamber **40**) and positioned such that inert gas (e.g., nitrogen gas) from these ports flow over the top surface of wafer **10** seated on, or proximate, cold plate **94**. In some embodiments, by combining convective and conductive heat transfer, wafer **10** can be cooled at a rate greater than or equal to ($\geq$) about 60° C./minute. Although not required, in some embodiments, the region from base of enclosure **66** (i.e., top surface of cold plate **94**) to a height of about 60 mm from base can form the cool-down zone **90A** of enclosure **66**. That is, in some embodiments, cool-down zone **90A** can be a region of enclosure spanning about 0-60 mm from the base of enclosure **66**.

[0042] Transfer zone **90B** is a region above the cool-down zone **90A**. The transfer zone **90B** can be aligned with port **42** (see FIGS. 1B and 5B) of chamber **40**. As explained previously, wafer **10** is inserted into enclosure **66** via port **42**. When wafer **10** is inserted into enclosure **66**, it enters enclosure **66** at transfer zone **90B**. When wafer **10** is inserted, spindle **44** is positioned in transfer zone **90B** such that the inserted wafer **10** can be supported on nubs **48A, 48B, 48C** of spindle **44**. Although not required, in some embodiments, transfer zone **90B** can be a region of enclosure spanning about 60-90 mm from the base of enclosure **66**.

[0043] Dosing zone **90C** is a region of the enclosure **66** above the transfer zone **90B**. When wafer **10** is positioned in the dosing zone **90C**, formic acid vapor (or another chemical vapor) from bubbler **300** is directed into the enclosure **66** of processing chamber **40** via outlet tube **350** (see

FIG. 5D) and gas port **42** (see FIG. 5B). This formic acid vapor can be directed to a chemical delivery tube **98** positioned proximate the top of enclosure **66**. The formic acid vapor from the chemical delivery tube **98** can be directed to the dosing zone **90C** of enclosure **66**. The formic acid vapor can be injected into enclosure **66** by nozzles on the chemical delivery tube **98** while the spindle **44** is rotating. The chemical vapor covers the top surface of wafer **10** and treats all regions of the wafer **10** substantially uniformly. In some embodiments, an edge ring directs chemical vapor to the top of the wafer. Although not required, in some embodiments, dosing zone **90C** can be a region of enclosure that spans about 90-120 mm from the base of enclosure **66**.

[0044] The fast ramp zone **90D** is a region of enclosure **66** above the dosing zone **90C**. When spindle **44** raises wafer **10** to fast ramp zone **90D**, lamps **82** on the underside of lid **70** heat wafer **10**. In some embodiments, lamps **82** can heat wafer **10** at a rate $\geq$ about 120° C./minute. Rotation of spindle **44** can ensure that all regions of the wafer **10** are heated substantially uniformly. One or more thermocouples and/or pyrometers can measure temperature of wafer **10** during heating. In some embodiments, control system **200** can control power of lamps **82** using a feedback loop (e.g., PID control) to control the wafer temperature. In some embodiments, one or more thermocouples can be provided in spindle **44** to measure the temperature of the wafer **10** during heating. In some such embodiments, a spring may keep thermocouple(s) in contact with wafer **10**. In some embodiments, one or more pyrometers focused on the rotating wafer **10** can be disposed at top or bottom of the enclosure **66** to measure the temperature of wafer **10** during heating.

[0045] As explained previously, when wafer **10** is positioned in dosing zone **90C**, formic acid vapor from the bubbler **300** is injected into enclosure **66** via chemical delivery tube **98** positioned at the top of enclosure **66**. Chemical delivery tube **98** can have a shape configured to discharge chemical vapor substantially uniformly over the wafer **10**. FIGS. 7A-7C illustrate different configurations of chemical delivery tubes **98A-98C** that can be used in different embodiments of oven **100**. In some embodiments, as illustrated in FIG. 7A, chemical delivery tube **98A** can be straight tube that extends from one side of chamber **40** to the center of wafer **10**. Multiple nozzles **99** or ports can be positioned on underside of tube **98A** facing wafer **10**. Any number (2-20) of nozzles **99** can be provided on tube **98A**. Dispersing gas into enclosure **66** at different radial locations of the rotating wafer **10** can assist in treating solder bumps in all areas of wafer **10** substantially uniformly. In some embodiments, as illustrated in FIG. 7B, an angled (or bent) chemical delivery tube **98B** can be used. In some embodiments, as illustrated in FIG. 7C, a T-shaped chemical delivery tube **98C** can be used. Although not shown in FIGS. 7B and 7C, spaced apart nozzles **99** can be provided on underside of tubes **98B**, **98C** facing wafer **10** to uniformly treat all regions of the rotating wafer **10**. In some embodiments, based on the type, size, and/or shape of the wafer **10** being processed, a different configuration of chemical delivery tube **98A**, **98B**, **98C** can be used.

[0046] In some embodiments, when formic acid vapor is directed into the enclosure **66** via chemical delivery tube **98** on the top side of wafer **10**, inert gas (e.g., nitrogen) can be directed into the enclosure **66** on the bottom side of wafer **10**. Inert gas can be introduced into enclosure through openings **95** on cold plate **94** and/or through gas ports **96**. Simultaneously directing the inert gas under wafer **10** when formic acid vapor is directed above wafer **10** allows the formic acid vapor to cover and treat all regions on the top surface of wafer substantially evenly.

[0047] An exemplary method **200** of processing a wafer **10** in oven **100** using the exemplary reflow profile of FIG. 3 will now be described. As illustrated in FIG. 3, reflow profile of FIG. 3 is preferably defined by six time-temperature process zones (zones 1-6). A wafer **10** is first loaded into enclosure **66** of chamber **40** through port **42** (step **210**). Wafer **10** can be inserted into enclosure **66** via port **42** and disposed on spindle **44** positioned in transfer zone **90B**, for example, using robotic arm. Spindle **44** can rotate wafer (step **220**) and raise wafer **10** to fast ramp zone **90D** of enclosure **66** (step **230**). In some embodiments, pressure in enclosure **66** can also be evacuated to a pressure below atmospheric pressure. In some embodiments, enclosure **66** can be pumped down to a pressure between about 100 milli Torr-100 Torr. In some embodiments, the enclosure can be

pumped down to a pressure between about 1-10 Torr. Control system **200** activates lamps **82** of lamp assembly **80** to heat wafer **10** and increase its temperature (step **240**). In some embodiments, lamps **82** may be turned on before wafer **10** is loaded on spindle **44** in step **210**. While in fast ramp zone **90D** (i.e., step **230**), temperature of wafer **10** can increase as illustrated in zone 1 of FIG. 3. [0048] When thermocouples and/or pyrometers of enclosure **66** indicate that temperature of wafer **10** has reached a target temperature of process zone 1 (about 150° C. in FIG. 3), control system **200** can lower spindle **44** with wafer **10** to dosing zone **90C** (step **250**). While in dosing zone **90C**, chemical vapor (e.g., formic acid vapor) can be directed into enclosure **66** through chemical delivery tube **98** above wafer **10** while simultaneously directing inert gas (e.g., nitrogen) into enclosure **66** through gas-inlet ports **96** (and/or openings **95** on cold plate **94**) below wafer **10** (step **260**). Inert gas flow below wafer **10** assists in keeping chemical vapor above wafer **10** such that substantially all regions of rotating wafer **10** are evenly exposed and coated with vapor. During this step, temperature of the wafer **10** can follow the path illustrated in zone 2 of FIG. 3.

[0049] After wafer **10** is coated with chemical vapor in step **260**, enclosure **66** can be evacuated using a vacuum pump to remove excess vapor in enclosure **66** (step **270**). Alternatively, or additionally, in some embodiments, inert gas (e.g., nitrogen) can be directed into enclosure **66** during this step to exhaust residual chemical vapor that has not been deposited on wafer. Spindle **44** can then raise wafer **10** back to fast ramp zone **90D** to continue heating wafer **10** at fast rate (step **280**). During this step, temperature of the wafer **10** can follow the profile illustrated in zones 3-5 of FIG. 3 and solder reflows.

[0050] After solder reflow is completed, lamps **82** are turned off (or reduced to a safe idle power) by control system **200**, and wafer **10** is lowered to the cool-down zone **90A** to cool substrate (step **290**). In cool-down zone **90A**, in some embodiments, wafer **10** can physically contact, or be placed in close proximity to, cold plate **94** such that bottom surface of wafer **10** is cooled conductively or radiatively through liquid cooled cold plate **94** and convectively from top via inert (e.g., nitrogen) gas from gas-inlet ports **96** flowing over top of wafer **10**. In some embodiments, inert gas (e.g., nitrogen) can also be directed in through openings **95** in cold plate **94** to assist in cooling back side of wafer **10**. While in cool-down zone **90A**, temperature of wafer **10** can follow profile illustrated in zone 6 of FIG. 3. After cooling is complete, spindle **44** can raise wafer **10** to transfer zone **90B** and wafer **10** can be removed from enclosure through port **42** (step **300**).

[0051] FIG. 9 illustrates another exemplary method **400** of performing high temperature processing (such as, for example, solder reflow) using oven **100**. Method **400** will be described with reference to the exemplary reflow profile of FIG. 3. However, method **400** can be applied to any reflow profile or any high temperature processing. FIG. 9 is a flow chart of the process flow of method **400**, and FIGS. 10 and 11 illustrate the process conditions in the enclosure **66** (of oven **100**) during the different steps of method **400**. FIG. 10 is a graph showing the temperature of the enclosure **66** (in degrees centigrade (° C.)), and the flow rate of formic acid vapor (in standard cubic centimeters per minute (SCCM or cm.sup.3/min)) into the enclosure **66** during different steps of method **400**. In FIG. 10, the X-axis shows the time in seconds, the left Y-axis shows the temperature in° C., and the right Y-axis shows the formic acid vapor flow rate in cm.sup.3/min during different steps of method **400**. In FIG. 11, the X-axis shows the time in seconds, the left Y-axis shows the pressure in enclosure **66** in Torr, and the right Y-axis shows the percentage oxygen concentration (in parts per million (ppm)) in enclosure **66** during the different steps of method **400**. In FIGS. 10 and 11, labels corresponding to the different steps of method **400** (of FIG. 9) are included to indicate the evolution of the process conditions in the enclosure **66** during the different steps. In the discussion below, reference will be made to FIGS. 9-11.

[0052] As described with reference to method **200** (of FIG. 8), a wafer **10** is first loaded into enclosure **66** of chamber **40** through port **42** and disposed on spindle **44** positioned in transfer zone **90B**. The spindle **44** can then be activated to rotate the wafer **10**. With reference to FIG. 9, in step **410**, the pumping system of oven **100** (e.g., vacuum pump, etc.) then pumps down enclosure **66** to

reduce the oxygen concentration in enclosure **66**. In some embodiments, as can be seen in FIG. **11**, in step **410**, the enclosure **66** can be cycled multiple times between about 0.1 Torr and about 50 Torr to reduce the oxygen concentration in enclosure **66**. In some embodiments, at the end of step **410**, the oxygen concentration may be below about 750 ppm or below about 500 ppm. U.S. Pat. No. 10,147,617, incorporated herein by reference in its entirety as if fully set forth herein, describes an exemplary process by which enclosure **66** can be cycled multiple times between a relatively higher and a relatively lower pressure to achieve a relatively oxygen-free process environment. [0053] In step **420**, spindle **44** raises wafer **10** to fast ramp zone **90D** of oven **100**. While in fast ramp zone **90D**, top surface of the wafer **10** is positioned proximate lamps **82** on underside of lid **70**. Lamps **82** increase the temperature of wafer **10**. Lamps **82** may be activated before or after the wafer **10** is moved to the fast ramp zone **90D**. In some embodiments, in step **420**, the lamps **82** can heat the wafer to a temperature between about 150° C.-180° C. in step **420**. As previously explained, this temperature range is exemplary and the wafer may be heated to any temperature (e.g., based on the reflow profile followed) in this step.

[0054] In step **430**, the heated wafer **10** is lowered to the dosing zone 90° C. of oven **100**. While in this zone, formic acid vapor from bubbler **300** is admitted into chamber through the chemical delivery tube **98** disposed at the top of enclosure **66** (see FIG. **6A**). This formic acid vapor is injected into enclosure **66** through the nozzles the chemical delivery tube **98**. Using bubbler **300** to produce formic acid vapor is only exemplary. In general, formic acid vapor may be produced by any appropriate source (such as, for example, vaporizer, etc.). Formic acid vapor may cover wafer surface by diffusion and forced convection. An edge ring of oven **100** can direct the chemical vapor to the top surface of the wafer **10**. The spindle **44** (or another wafer support mechanism) can rotate the wafer **10** for uniform chemical vapor coating on the surface of the wafer **10**. The pressure of the enclosure above the wafer **10** (or the entire enclosure **66**) in step **430** may be below about 20 Torr. In some embodiments, the pressure can be below about 10 Torr. In some embodiments, the pressure can be between about 1-8 Torr. [done].

[0055] In some embodiments, nitrogen gas (or another inert gas) can also be discharged into the enclosure **66** below the wafer **10** to keep the formic acid vapor on top of the enclosure **66** to coat the top surface of the wafer **10** completely. Nitrogen can be discharged below the wafer **10** through the openings **95** (see FIG. **7A**) on cold plate **94** and/or through gas ports **96** on the side wall of the enclosure **66**. In some embodiments, the temperature in this zone (dosing zone 90° C.) can be controlled to be between 150° C.-180° C. (or any other temperature, for example, based on the reflow profile). Due to oxidation, the surface of the solder on wafer **10** may have oxides formed thereon. The formic acid vapor discharged into enclosure **66** through chemical delivery tube **98** can have a controlled concentration of formic acid configured for effective cleaning of these oxides.

[0056] After wafer **10** is coated with formic acid vapor in step **430**, the formic acid vapor flow into enclosure **66** is stopped. In step **440**, at least a portion of the excess formic acid vapor is removed or pumped out from enclosure **66** using the vacuum pump of oven **100**. A pump and purge process can be used to remove the excess formic acid vapor from enclosure **66**. In this step, the enclosure **66** is cyclically pumped down to a low pressure and purged with nitrogen to remove excess formic acid vapor from enclosure **66**. The wafer **10** remains in the dosing zone **90C** and the temperature of the wafer **10** is maintained between about 150° C.-180° C. during this step. In some embodiments, the concentration of formic acid vapor in the enclosure after the vapor removal process of step **430** can be below about 50%~80%. In some embodiments, the pressure in enclosure after the vapor removal process of step **430** can be between about 1-10 Torr. In some embodiments, in this step, the nitrogen flow rate into the chamber below the wafer can vary from about 100 sccm to about 5000 sccm to remove the excess formic acid from enclosure **66**.

[0057] In step **450**, the lamps **82** are controlled to increase the temperature of the wafer **10** (or the enclosure **66**) to a temperature between about 220° C.-250° C. for the soldering process. As previously explained, the lamps **82** can be controlled by control system **200** (FIG. **1A**) to change

the temperature of the wafer **10**. For example, the control system **200** can turn on/off a different number of lamps **82** and/or increase/decrease power to the lamps **82** to change the temperature of the enclosure **66**. In step **460**, the wafer **10** is maintained at a temperature between about 220° C.-250° C. for the soldering process. As previously explained, in general, the wafer can be heated to any temperature in this step. In some embodiments, the temperature that the wafer is heated to in step **450** can be higher than the temperature in step **420**. In some embodiments, the wafer **10** may be soaked or maintained at the temperature between about 220° C.-250° C. for between about 0.5-4 minutes, or between about 1-2 minutes. It should be noted that this soak time is only exemplary and may depend on the solder reflow profile used and the type of processing being carried out in oven **100**. In some embodiments, nitrogen flow below the wafer **10** can be continued in steps **450** and **460**. The temperature to which the wafer is heated in steps **450** and **460** depends on the reflow profile used. For example, if wafer **10** includes a higher melting temperature solder (such as, for example, gold-tin), a different reflow profile (than FIG. 3) can be used, and the wafer can be heated and maintained at a higher temperature to reflow the solder in steps **450** and **460**

[0058] In step **470**, after the soldering is completed, lamps **82** are turned off (or reduced to a safe idle power), and the wafer **10** is lowered into the cool-down zone **90A** of enclosure **66** to cool down the wafer. In some embodiments, the wafer **10** can be physically supported by the cold plate **94** to cool the wafer by conduction. In some embodiments, a liquid coolant can also be circulated through the coolant channels of cold plate **94** to quickly cool the wafer **10**. configured to direct a coolant (water or another liquid coolant) therethrough to remove heat conducted from wafer **10**. In some embodiments, the underside of the wafer **10** can be positioned proximate the cold plate **94** and nitrogen gas directed to the underside of the wafer via openings **95** (see FIG. 7A) on the cold plate **94** to cool underside of the wafer **10** by forced convection. In some embodiments, when wafer **10** is positioned in the cool-down zone **90A** (e.g., seated on or positioned proximate cold plate **94**), nitrogen gas (or other suitable inert gas) can be directed into enclosure **66** (e.g., via gas ports **96** on side wall of base **60** of chamber **40**) to flow above the wafer **10** and cool the top surface of wafer **10** by convection. In some embodiments, the wafer **10** can be cooled to a temperature between about 15-30° C. in the cool-down zone **90A**. In some embodiments, the wafer **10** can be cooled to about the room temperature in the cool-down zone **90A**. The cooled wafer **10** can then be raised to the transfer zone **90B** of enclosure **66** and removed from the chamber **40** of oven **100**.

[0059] For wafers and semiconductor devices using small solder bumps (or micro-bumps), and in 3D stacking of semiconductor devices, good coverage of formic acid vapor on the wafer and solder surface can promote effective removal of the surface oxide layers from the solder. A low pressure (i.e., below about 10 Torr) of the enclosure **66** when formic acid vapor is introduced into the enclosure **66** enables the vapor to effectively cover the top surface of the wafer **10** and clean the solder surface. Discharging nitrogen gas (or another inert gas) below the wafer **10** while formic acid vapor is discharged from the top also assists in confining the formic acid vapor primarily on the top surface of the wafer **10**. The concentration of the formic acid in the vapor discharged into the enclosure **66** (in step **430**) is also controlled for effective cleaning of the surface oxides on the solder. Controlling the diffusion coefficient (or diffusivity) of the formic acid vapor discharged into the enclosure **66** in step **430** can improve oxide cleaning. The diffusivity of the formic acid vapor is improved by controlling the formic acid concentration and the resultant partial pressure of formic acid and temperature dependent gas mobility of the vapor to improve the soak process in step **430**. In some embodiments of the current disclosure, the concentration of the formic acid in the formic acid vapor discharged into the enclosure in step **430** can be between about 3-15%. In some embodiments, the formic acid concentration in the vapor can be between about 30-40%. This formic acid concentration can result in the diffusivity of the formic acid vapor to be between about 200-300 cm.sup.2/second, and a partial pressure between about 20-40%.

[0060] The above-described embodiment of reflow oven **100** and methods of operating the oven are only exemplary. Many variations are possible. Although formic acid vapor is described as the

chemical vapor discharged into oven **100** (in step **430**), this is only exemplary. Any chemical vapor suitable for the process being carried out in oven **100** can be used. Further, any reflow process can be performed using oven **100** using a time-temperature profile (or reflow profile) suitable for solder material being used. U.S. Patent Application Publication US 2021/0265301, incorporated by reference as if fully set forth herein, describes some exemplary reflow processes that can be performed in oven **100**. The steps of methods **200** and **400** need not be performed in the order illustrated in FIGS. **8** and **9**. Some steps can be omitted and/or some steps added in other exemplary methods. For example, in some embodiments, lamps **82** can be activated (step **240**) before loading wafer into the chamber (i.e., step **210**) or before rotating spindle (step **220**), etc. As another example, in some embodiments, enclosure **66** of chamber **40** can be pumped down to a sub-atmospheric pressure (e.g., between about 1-10 Torr) loading the wafer into chamber (step **210**), etc. Persons of ordinary skill in the art would recognize and understand these possible variations to be within the scope of the present disclosure.

[0061] Furthermore, although oven **100** is disclosed in conjunction with a solder reflow process, this is only exemplary. Persons of ordinary skill in the art would recognize that the disclosed oven can be used for any appropriate process. Further, although a wafer bumping process is described, this disclosure is not limited to this application. In general, any high temperature process may be performed in oven **100** on any type of substrate (e.g., wafer, organic/ceramic substrates, semiconductor packages, printed circuit board (PCB), etc.). For example, in some embodiments, oven **100** can be used for attaching dies (or IC chips) on substrates or chip-carriers (PCB, ceramic/organic substrates, etc.) using solder (e.g., reflow soldering). During reflow soldering, solder balls or solder paste (a sticky mixture of powdered solder and flux) can be used to temporarily attach one or more electronic components together (e.g., dies to substrates using C4 joints, die-substrate assembly to package substrate (e.g., PCB) using ball grid array (BGA) joints, etc.). The component assembly can then be subject to a reflow process to melt the solder and attach the components together.

[0062] Other embodiments of the oven and related methods will be apparent to those skilled in the art from consideration of the disclosure herein. It should also be noted that, in some embodiments, the functions of one or more of the described zones of the oven **100** (e.g., cool-down zone **90A** transfer zone **90B** directly above cool-down zone **90A**, dosing zone **90° C.** fast ramp zone **90D**, etc.) may be combined, thereby reducing the number of zones.

Claims

21. A method of using a processing oven, comprising: disposing at least one substrate on a rotatable spindle positioned in a chamber of the oven; rotating the spindle and the at least one substrate about a central axis of the chamber; increasing a temperature of the at least one substrate to a first temperature while the at least one substrate is rotating on the spindle in the chamber; after increasing the temperature, admitting a chemical vapor into the chamber while the at least one substrate is rotating on the spindle in the chamber; after admitting the chemical vapor, further increasing the temperature of the at least one substrate to a second temperature higher than the first temperature while the at least one substrate is rotating on the spindle in the chamber; and further increasing the temperature, cooling the at least one substrate in the chamber.

22. The method of claim 21, wherein increasing the temperature to the first temperature includes activating a heating assembly disposed above the at least one substrate.

23. The method of claim 22, wherein the heating assembly is a lamp assembly.

24. The method of claim 21, wherein increasing the temperature to the first temperature includes increasing the temperature of the at least one substrate to a temperature between about 150° C.-180° C.

25. The method of claim 24, wherein further increasing the temperature includes further increasing

the temperature of the at least one substrate to a temperature between about 220° C.-250° C.

26. The method of claim 21, wherein admitting the chemical vapor into the chamber includes admitting the chemical vapor into the chamber above the at least one substrate.

27. The method of claim 21, wherein admitting the chemical vapor into the chamber includes admitting formic acid vapor into the chamber.

28. The method of claim 27, wherein admitting formic acid vapor into the chamber includes directing formic acid vapor into the chamber from a source configured to vary a concentration of formic acid in the formic acid vapor.

29. The method of claim 21, wherein cooling the at least one substrate includes cooling the at least one substrate to room temperature.

30. The method of claim 21, wherein cooling the at least one substrate includes contacting the at least one substrate with a cold plate of the chamber.

31. The method of claim 30, further including directing a liquid coolant through the cold plate to assist in cooling the at least one substrate.

32. The method of claim 21, wherein cooling the at least one substrate includes directing an inert gas on the at least one substrate to assist in cooling the at least one substrate.

33. A method of using a processing oven, comprising: supporting at least one substrate on a rotatable spindle positioned in a chamber of the processing oven; rotating the spindle with the at least one substrate in the chamber; heating the at least one substrate to a first temperature using a heating assembly of the chamber disposed above the at least one substrate while the at least one substrate is rotating on the spindle in the chamber; directing a chemical vapor into the chamber; after directing the chemical vapor, further heating the at least one substrate to a second temperature higher than the first temperature by moving the spindle towards the heating assembly; and after the further heating, cooling the at least substrate in the chamber to a temperature below the first temperature.

34. The method of claim 33, wherein directing the chemical vapor into the chamber includes directing formic acid vapor into the chamber.

35. The method of claim 33, wherein the at least one substrate includes a solder thereon, and wherein the first temperature is a temperature below a reflow temperature of the solder and the second temperature is a temperature above the reflow temperature.

36. A method of using a processing oven, comprising: disposing at least one substrate on a rotatable spindle positioned in a chamber of the oven, wherein the at least one substrate includes a solder thereon; rotating the spindle and the at least one substrate about a central axis of the chamber; increasing a temperature of the at least one substrate to a temperature below a reflow temperature of the solder while the at least one substrate is rotating on the spindle in the chamber; admitting formic acid vapor into the chamber; after admitting the chemical vapor, further increasing the temperature of the at least one substrate to a temperature higher than the reflow temperature of the solder while the at least one substrate is rotating on the spindle in the chamber; further increasing the temperature, cooling the at least one substrate in the chamber.

37. The method of claim 36, further including activating a heating assembly disposed above the at least one substrate.

38. The method of claim 37, wherein the heating assembly is a lamp assembly.

39. The method of claim 36, wherein further increasing the temperature of the at least one substrate includes heating the at least one substrate at a ramp rate greater than or equal to 120° C./minute.

40. The method of claim 36, wherein cooling the at least one substrate includes cooling the at least one substrate at a cooling rate greater than or equal to about 60° C./min.
